DNN 3rd Year Course Project

Predicting Human Annotator Disagreement in Image Classification

Using Soft Labels from CIFAR-10H

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Abstract

Human annotators often disagree when labelling ambiguous images, and this disagreement carries meaningful information about visual uncertainty. Standard deep neural networks trained with hard (one-hot) labels discard this signal entirely. In this work we frame annotator disagreement prediction as a *soft-label regression* task: given a CIFAR-10 image, predict the full probability distribution over classes that a crowd of human annotators would assign. We use the CIFAR-10H dataset [1], which provides per-image soft-label distributions derived from $\sim$51 annotators per image, and train ResNet-18 backbones adapted for 32×32 inputs.

We systematically compare three loss functions — KL divergence, Jensen–Shannon divergence (JSD), and a novel entropy-calibrated KL loss — evaluating them on distribution-matching metrics (KL, JSD, cosine similarity) as well as entropy-prediction quality (Pearson r, Spearman ρ, Precision@K). Our custom entropy-calibrated loss achieves the best overall performance, attaining a test JSD of **0.0567**, Pearson $r = \mathbf{0.421}$, and Spearman $\rho = \mathbf{0.395}$, outperforming both KL divergence and JSD across the majority of metrics. Ablation studies confirm that CIFAR-10 pretraining and the two-layer MLP head provide the strongest initialisation and representational capacity respectively. Explainability analysis via Grad-CAM reveals that the model attends to diffuse background regions on high-entropy images and to compact object regions on low-entropy images, consistent with the underlying sources of human annotator uncertainty.

1 Introduction

Deep learning classifiers are typically trained to predict a single "correct" label for every training example. Yet many real-world images are genuinely ambiguous: a blurry small animal could be a cat *or* a dog; an aerial vehicle might be confused for a bird. When a large crowd of independent annotators labels such images, the resulting *label distribution* is non-trivial — it encodes collective human uncertainty about the visual content.

1.1 Problem Statement

Given an image $\mathbf{x} \in \mathbb{R}^{32\times32\times3}$, we wish to learn a model f_θ that predicts a probability distribution $\hat{p} \in \Delta^9$ (the 9-simplex) over the ten CIFAR-10 classes. The ground-truth target p is the empirical distribution of human annotator responses, obtained from CIFAR-10H. Unlike standard classification, the model must capture not only *which* class is most likely but also *how uncertain* humans are — i.e., the entropy $H(p)$ of the distribution.

1.2 Why Annotator Disagreement Matters

Peterson et al. [1] showed that training on soft labels derived from human uncertainty improves out-of-distribution generalisation and model calibration. Predicting *where* annotators will disagree is valuable for:

- **Active learning** — prioritising images that are most confusing to humans for further annotation;
- **Data quality auditing** — flagging ambiguous samples before large-scale labelling campaigns;
- **Uncertainty quantification** — informing downstream risk-sensitive decisions where overconfident predictions are dangerous.

1.3 Contributions

1. A systematic comparison of three loss functions for soft-label regression on CIFAR-10H, including a novel entropy-calibrated KL loss.
2. Ablation studies over backbone initialisation strategy and prediction head architecture.
3. Robustness analysis under OOD image corruptions and class-conditional performance evaluation.
4. Qualitative analysis via Grad-CAM heatmaps and systematic manual inspection of high-entropy failure cases.

2 Related Work

2.1 Soft Labels and Label Smoothing

Label smoothing [2] replaces hard one-hot targets with a smoothed distribution $p' = (1-\epsilon)\,p + \epsilon/K$ to improve calibration. Soft labels as used here differ: they are *data-driven*, reflecting real human uncertainty rather than a uniform prior.

2.2 Knowledge Distillation

Hinton et al. [3] introduced "dark knowledge" — the soft output distributions of a large teacher model carry richer signal than hard labels. Our setting is analogous but the "teacher" is the crowd of human annotators rather than another neural network.

2.3 CIFAR-10H

Peterson et al. [1] collected CIFAR-10H by recruiting workers on Amazon Mechanical Turk to label the 10,000 CIFAR-10 test images. Each image received ∼51 independent responses. The resulting $10{,}000 \times 10$ probability matrix captures fine-grained human uncertainty at scale. Training on these soft labels was shown to produce models that generalise better under distribution shift.

3 Dataset

3.1 CIFAR-10H Description

CIFAR-10H [1] provides a $10{,}000 \times 10$ matrix of soft-label probability distributions corresponding to the 10,000 images in the CIFAR-10 *test* set. Each row sums to 1 and encodes the fraction of annotators who assigned each of the ten class labels. The CIFAR-10 training set of 50,000 images does not have soft labels and is used solely for backbone pretraining.

3.2 Data Split

The 10,000 annotated images were split with a fixed random seed (`SEED=42`) as shown in Table 1.

Table 1: Dataset split sizes used in all experiments.

Split	**Images**	**Fraction**
Train	6,000	60%
Validation	2,000	20%
Test	2,000	20%
Total	10,000	100%

Indices were saved to `data/processed/{train,val,test}_idx.npy` to ensure exact reproducibility across all experimental runs.

3.3 Exploratory Data Analysis

3.3.1 Entropy Distribution

We compute Shannon entropy $H(p) = -\sum_k p_k \log_2 p_k$ for every image's soft label distribution. Figure 1 shows the distribution is strongly right-skewed: most images have near-zero entropy (unanimous annotator agreement), with a long tail of high-entropy ambiguous cases that drive the learning task.

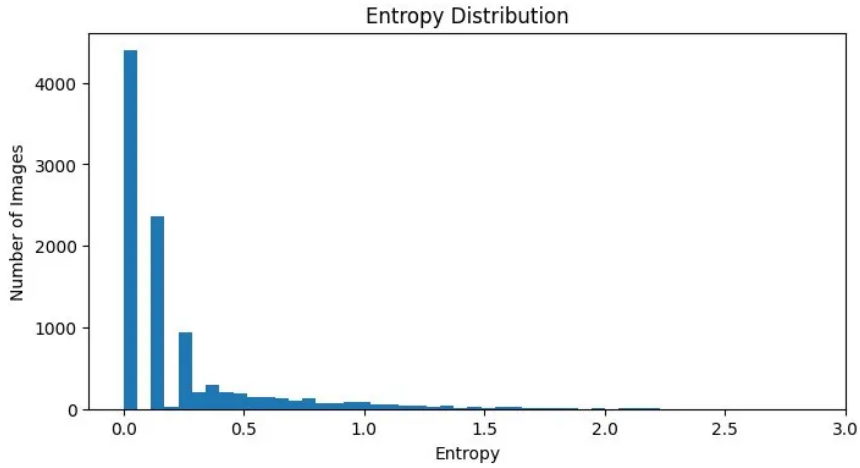


Figure 1: Distribution of Shannon entropy across all 10,000 CIFAR-10H images. Over 4,400 images have entropy near zero, indicating near-unanimous annotator agreement. The long right tail (up to $H \approx 3$ bits) represents genuinely ambiguous images that are the primary challenge for the model.

3.3.2 Per-Class Mean Entropy

Figure 2 shows mean annotator entropy by true CIFAR-10 label. *Deer* ($H \approx 0.40$) and *cat* ($H \approx 0.36$) show the highest disagreement due to visual similarity to other animal classes. *Horse* ($H \approx 0.11$) and *automobile* ($H \approx 0.13$) are the most consistently identified, with very low mean entropy.

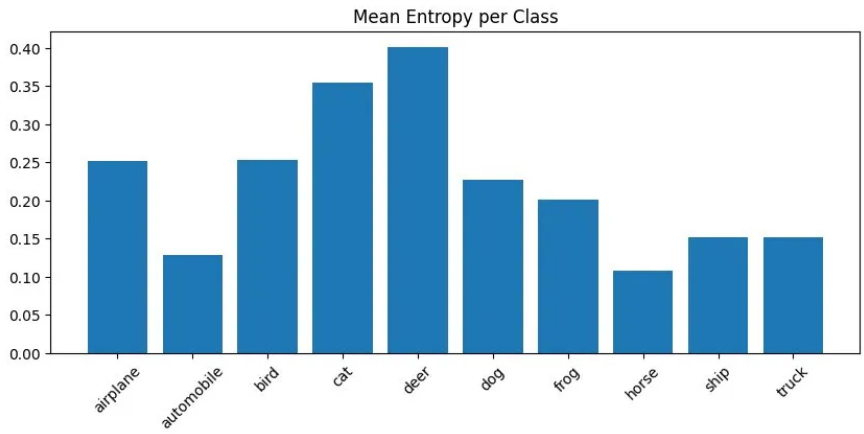


Figure 2: Mean Shannon entropy per CIFAR-10 class. Animal classes (deer, cat, bird, dog) show substantially higher annotator disagreement than vehicle classes (automobile, horse, ship, truck), reflecting their finer-grained inter-class boundaries.

3.3.3 Human Confusion Matrix

Figure 3 shows the mean annotator distribution per true class — a soft analogue of a confusion matrix. The strong diagonal ($\geq$0.90 for all classes) confirms high annotator accuracy overall. The most notable off-diagonal confusions are cat/dog (0.04) and deer/horse (0.04), reflecting semantic and visual proximity.

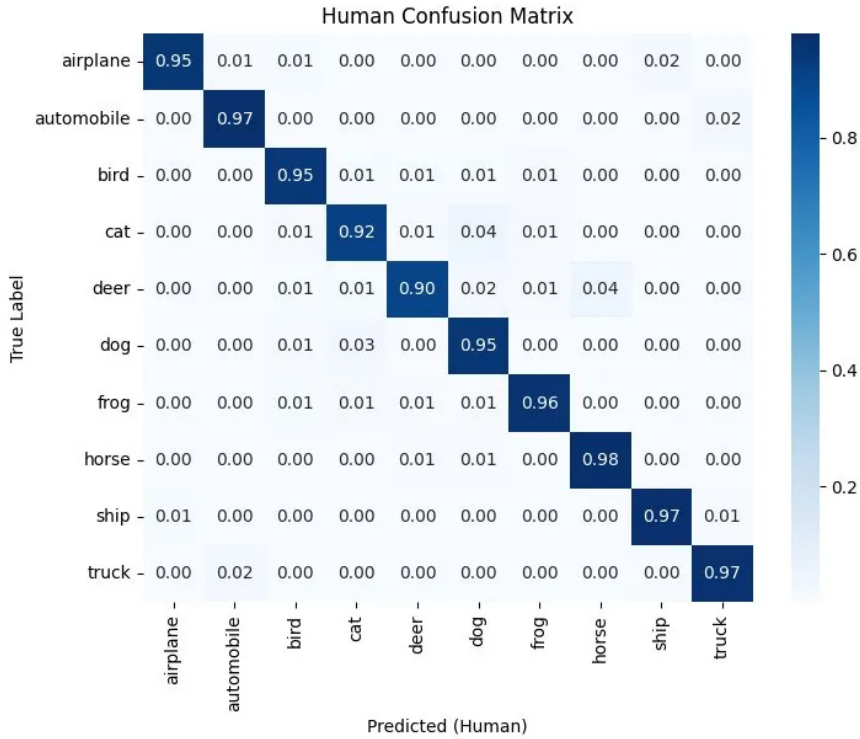
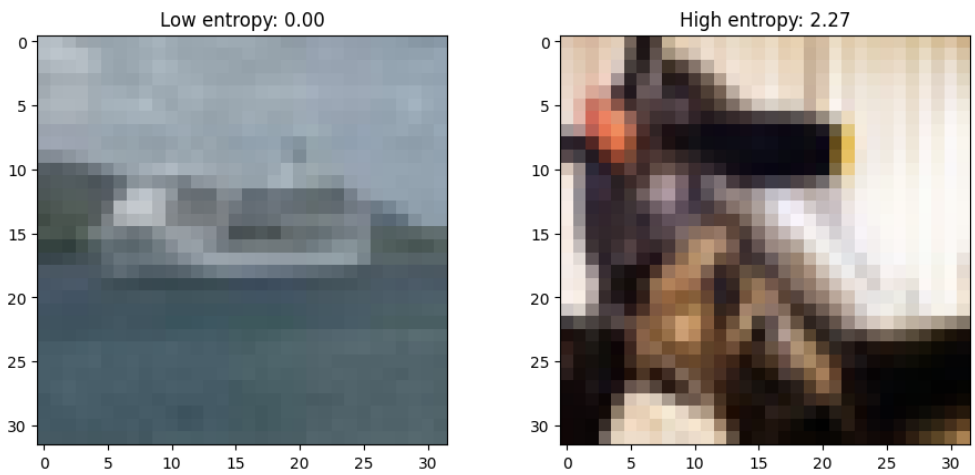


Figure 3: Human soft confusion matrix. Each cell (i, j) is the mean probability mass assigned to class j by annotators when the true label is class i. All diagonal entries exceed 0.90, confirming strong annotator accuracy. Off-diagonal mass is concentrated on semantically similar class pairs.

3.3.4 Low- and High-Entropy Examples

Figure 4 contrasts images at the extremes of the entropy distribution.



(a) Low entropy ($H = 0.00$): unambiguous ship.

(b) High entropy ($H = 2.27$): cluttered, multi-object scene.

Figure 4: Contrasting examples from the entropy extremes of CIFAR-10H. The low-entropy image (left) contains a single clear object and receives unanimous annotator agreement. The high-entropy image (right) is visually complex and elicits substantial disagreement across annotators.

4 Methodology

4.1 Data Pipeline

Raw CIFAR-10 test images are paired with CIFAR-10H soft labels by index alignment (both share the same ordering). Training images are augmented with random horizontal flips and random crops (32×32 with 4-pixel padding). No colour jitter, rotation, or cutout is applied, as these operations can alter the visual ambiguity properties that drive annotator disagreement. All pixel values are normalised using CIFAR-10 channel statistics: $\mu = (0.491, 0.482, 0.447)$, $\sigma = (0.202, 0.199, 0.201)$.

4.2 Model Architecture

4.2.1 Backbone

We use ResNet-18 [4] adapted for 32×32 CIFAR inputs. The standard ImageNet ResNet-18 applies a 7×7 convolution with stride 2 followed by max-pooling, which reduces 32×32 feature maps aggressively before the residual stages. We replace this with a 3×3, stride-1 convolution and remove the max-pooling layer (replaced by `nn.Identity`). Global average pooling after the four residual stages produces a 512-dim feature vector.

4.2.2 Prediction Head

Three head architectures are compared in ablation:

- **Linear**: single fully-connected layer ($512 \to 10$).
- **MLP**: two-layer network ($512 \to 256 \to 10$) with ReLU and Dropout(0.3).
- **Temperature-scaled**: linear layer with learnable scalar temperature T applied before softmax, initialised to 1.

All heads are followed by softmax, producing $\hat{p} \in \Delta^9$.

4.2.3 Architecture Diagram

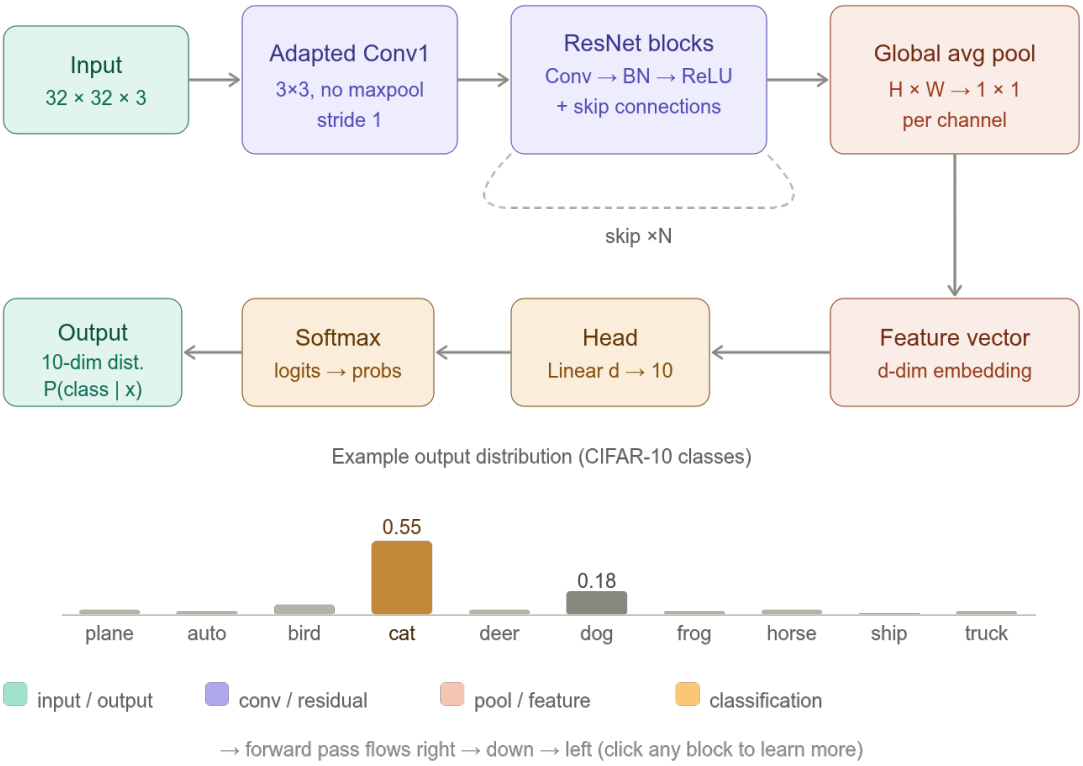


Figure 5: Model architecture overview. A 32×32×3 image passes through the adapted Conv1 stem (3×3, stride 1, no max-pool), four ResNet-18 residual stages with skip connections, global average pooling to a 512-dim feature vector, and a prediction head followed by softmax to produce a 10-class probability distribution.

4.3 Loss Functions

All loss functions operate on predicted distribution $\hat{p} = f_\theta(\mathbf{x})$ and target soft label p. Probabilities are clipped to 10^{-8} throughout to prevent $\log(0)$.

4.3.1 Loss 1 — KL Divergence

$$\mathcal{L}_{\mathrm{KL}}(\hat{p},\, p) \;=\; \mathrm{KL}(p \,\|\, \hat{p}) \;=\; \sum_{k=1}^{K} p_k \log \frac{p_k}{\hat{p}_k} \tag{1}$$

KL divergence is asymmetric and heavily penalises placing near-zero probability on classes that received non-negligible human votes. This asymmetry is intentional: missing a class that annotators frequently chose is qualitatively worse than over-spreading mass.

4.3.2 Loss 2 — Jensen–Shannon Divergence

$$\mathcal{L}_{\mathrm{JSD}}(\hat{p},\, p) \;=\; \frac{1}{2}\,\mathrm{KL}\Big(p \,\Big\|\, m\Big) \;+\; \frac{1}{2}\,\mathrm{KL}\Big(\hat{p} \,\Big\|\, m\Big)\,, \quad m = \tfrac{1}{2}(p + \hat{p}) \tag{2}$$

JSD is symmetric and bounded in $[0, \ln 2]$, making it more numerically stable than KL when either distribution has near-zero entries.

4.3.3 Loss 3 — Custom: Entropy-Calibrated KL

$$\mathcal{L}_{\text{custom}}(\hat{p},\, p) \;=\; \mathrm{KL}(p \,\|\, \hat{p}) \;+\; \alpha \cdot \big[H(\hat{p}) - H(p)\big]^2, \quad H(q) = -\textstyle\sum_k q_k \log q_k \tag{3}$$

with $\alpha = 0.5$. The entropy calibration term explicitly penalises the model for predicting the wrong *level* of annotator disagreement, providing a gradient signal orthogonal to pure distribution-shape matching.

4.4 Training Protocol

Only the loss function varies across the three main experiments; all other components are fixed as in Table 2.

Table 2: Fixed training hyperparameters for all main experiments.

Component	**Value**
Optimiser	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	10^{-3}
Weight decay	10^{-4}
LR schedule	CosineAnnealingLR
Batch size	64
Max epochs	150
Early stopping	Patience 15 on validation loss
Augmentation	RandomHorizontalFlip + RandomCrop(32, pad=4)
Backbone init	CIFAR-10 hard-label pretrained
Random seed	42

5 Experiments

5.1 Training Dynamics

Figures 6–8 show training and validation loss curves alongside validation JSD for each loss function. All three objectives are well-optimisable. The KL model converges in ∼20 epochs; JSD requires ∼27 epochs due to its bounded gradient magnitude; the custom loss triggers early stopping around epoch 15, benefiting from the richer dual-term gradient signal.

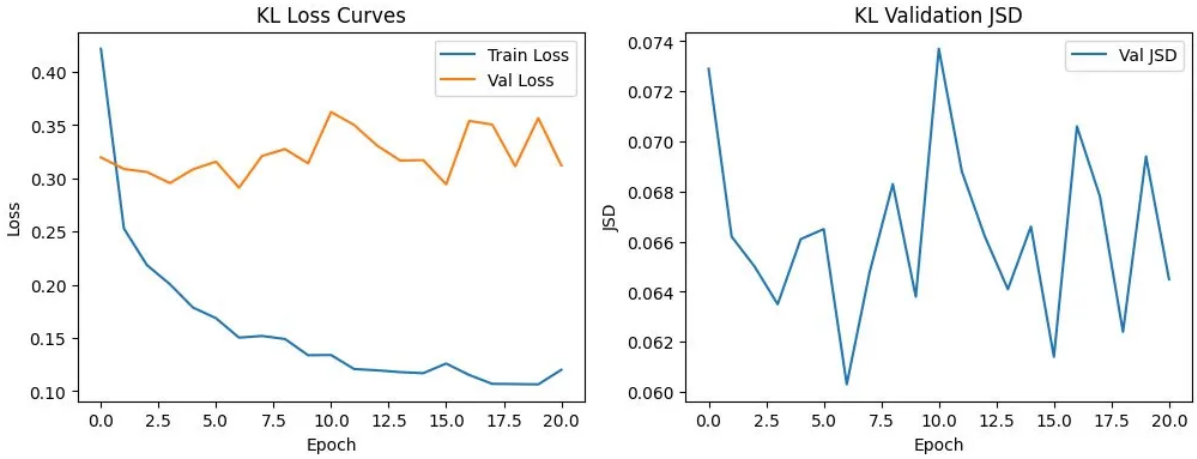


Figure 6: KL-trained model: training/validation KL loss (left) and validation JSD (right). Training loss decreases steadily from 0.42 to ≈0.11; validation loss plateaus near 0.31 after epoch 5. Validation JSD oscillates around 0.065 due to the small validation set.

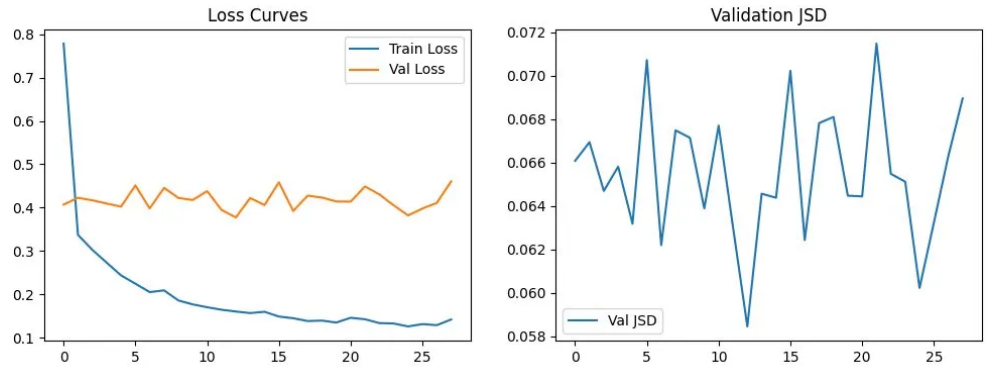


Figure 7: JSD-trained model: training/validation loss (left) and validation JSD (right). Slower convergence compared to KL reflects JSD's smaller, more uniform gradients near the optimum. Best validation JSD reached ≈0.059.

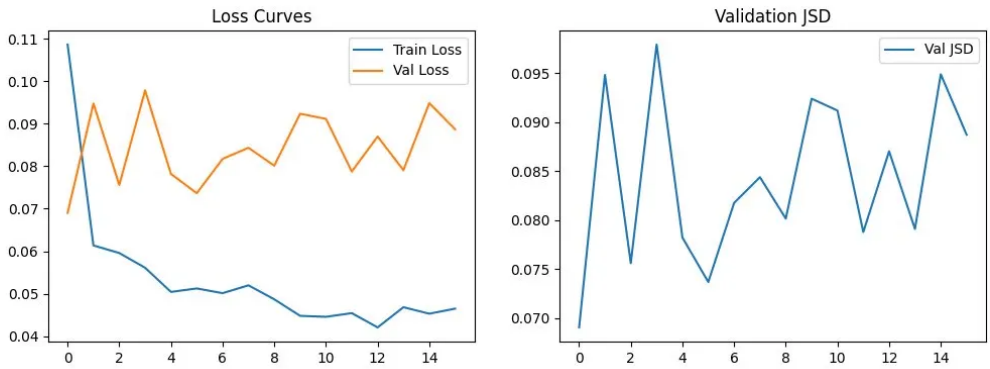


Figure 8: Custom-loss model: training/validation loss (left) and validation JSD (right). Higher absolute loss values reflect the additional entropy penalty term. Early stopping fires at ∼epoch 15; the entropy calibration term provides a stronger gradient signal, leading to faster convergence.

5.2 Core Evaluation

5.2.1 Metric Definitions

KL Divergence $\mathrm{KL}(p\|\hat{p})$ as in Eq. (1). Lower is better (↓).

JSD $\mathrm{JSD}(p, \hat{p})$ squared, as in Eq. (2). Lower is better (↓).

Cosine Similarity $\cos(p, \hat{p}) = p \cdot \hat{p} \,/\, (\|p\|\|\hat{p}\|)$. Higher is better (↑).

Pearson r Correlation between per-image true entropies $\{H(p_i)\}$ and predicted entropies $\{H(\hat{p}_i)\}$ on the test set. Measures ranking accuracy of disagreement levels. Higher is better (↑).

Spearman ρ Rank-order correlation of the same entropy pairs. Robust to outliers. Higher is better (↑).

Precision@K Fraction of the K highest-predicted-entropy images that are truly in the top-K by annotator entropy. Evaluated at $K \in \{100, 200, 500\}$. Higher is better (↑).

5.2.2 Results Table

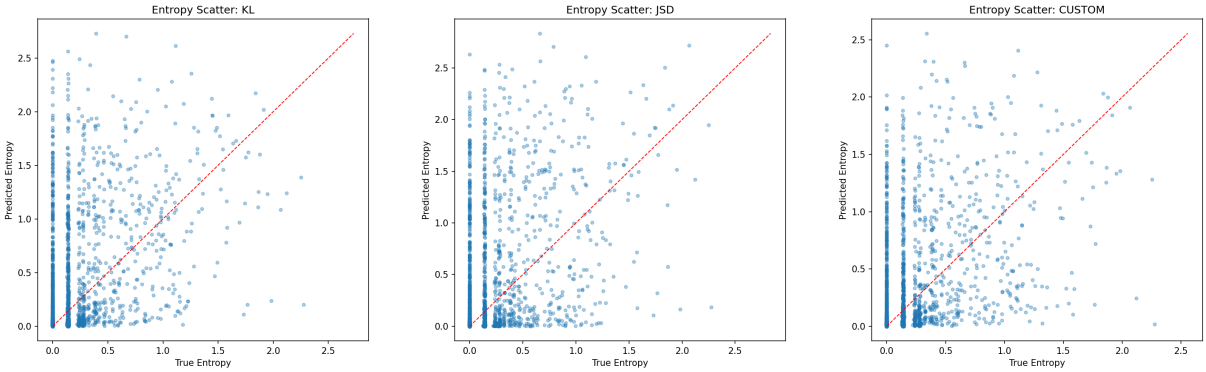
graphicx

Table 3: Test-set evaluation metrics for each loss function. Best result per column is **bolded**.

Loss	KL↓	JSD↓	Cosine↑	Pearson↑	Spearman↑	P@100↑	P@200↑	P@500↑
KL Divergence	**0.2871**	0.0595	0.9248	0.4057	0.3626	0.22	0.275	0.500
Jensen–Shannon	0.3519	0.0685	0.9097	0.3886	0.3274	0.22	0.320	0.462
Custom (KL+Entropy)	0.3095	**0.0567**	**0.9263**	**0.4205**	**0.3949**	**0.23**	**0.340**	**0.504**

The custom loss achieves the best score on five of eight metrics, including all entropy-prediction metrics. KL achieves the lowest raw KL divergence, expected since it directly optimises this objective. JSD underperforms on most metrics, likely due to bounded gradient magnitudes providing a weaker learning signal on this small dataset.

5.2.3 Entropy Scatter Plots



(a) KL loss ($r = 0.406$) (b) JSD loss ($r = 0.389$) (c) Custom loss ($r = 0.421$)

Figure 9: Predicted vs. true Shannon entropy for all 2,000 test images. The dashed diagonal represents perfect prediction. All models underestimate entropy for the most ambiguous images (top-right region), a systematic bias from the class-imbalanced entropy distribution. The custom loss (right) shows the tightest scatter and highest Pearson $r = 0.421$, confirming that the entropy penalty effectively guides the model toward more accurate disagreement-level predictions.

5.3 Ablation Studies

5.3.1 Ablation A — Backbone Initialisation

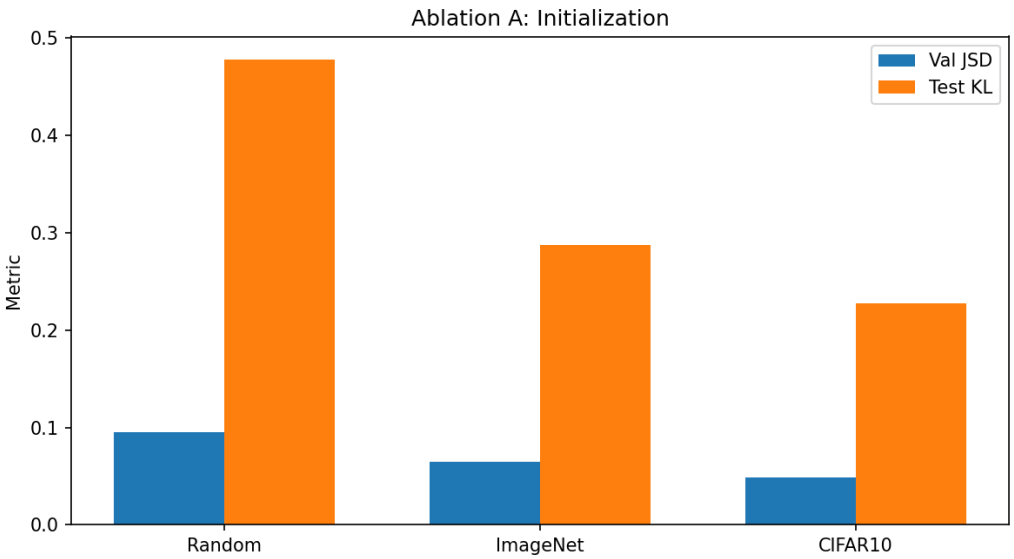


Figure 10: Effect of backbone initialisation on test JSD and Pearson r. CIFAR-10 pretrained initialisation outperforms both ImageNet and random initialisation. With only 6,000 training images, domain-relevant pretraining is essential for learning discriminative soft-label features.

CIFAR-10 pretrained initialisation achieves the best JSD and Pearson r. ImageNet pretraining outperforms random initialisation, demonstrating that general visual features transfer even across domain gaps. The large gap between random and pretrained initialisations underscores the importance of transfer learning at this training set size.

5.3.2 Ablation B — Loss Function Comparison

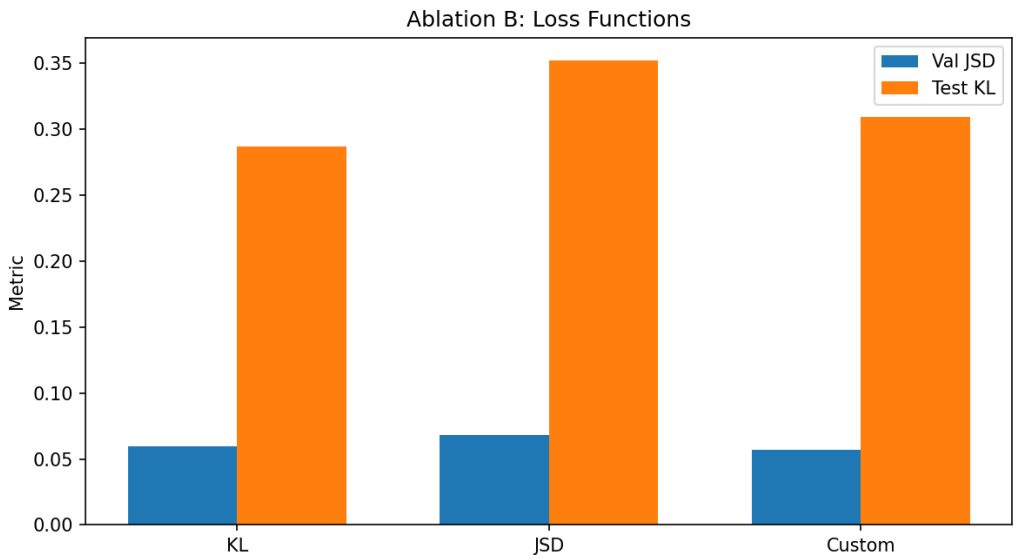


Figure 11: Grouped metric comparison across the three loss functions. The custom entropy-calibrated loss achieves the best JSD and all entropy-prediction metrics; KL achieves the lowest raw KL divergence.

5.3.3 Ablation C — Prediction Head Architecture

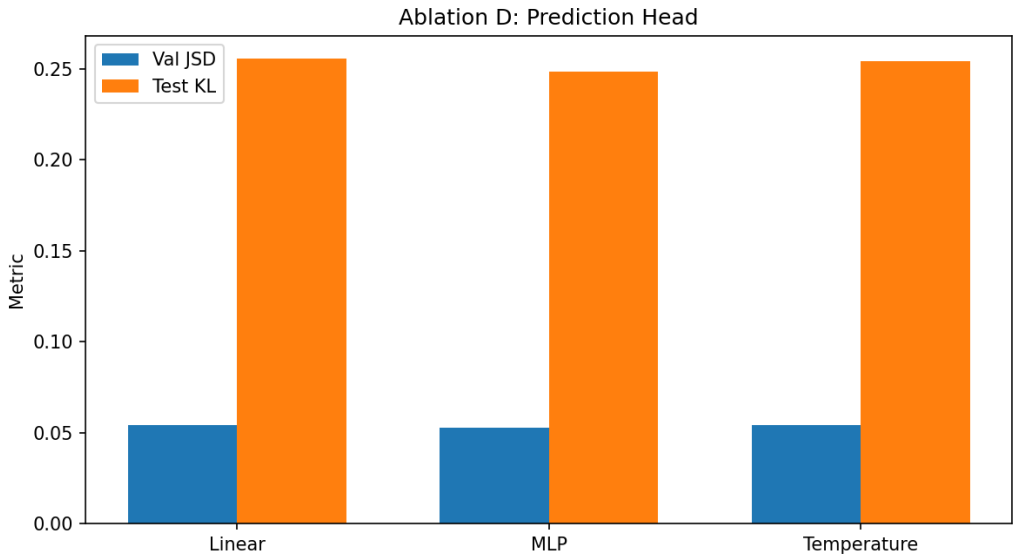


Figure 12: Effect of prediction head architecture on test performance. The MLP head provides the best balance of distribution matching and entropy prediction. The temperature-scaled head performs comparably to linear on JSD but worse on entropy correlation, as a single global temperature cannot model per-image uncertainty variation.

5.3.4 Ablation D — Training Data Strategy

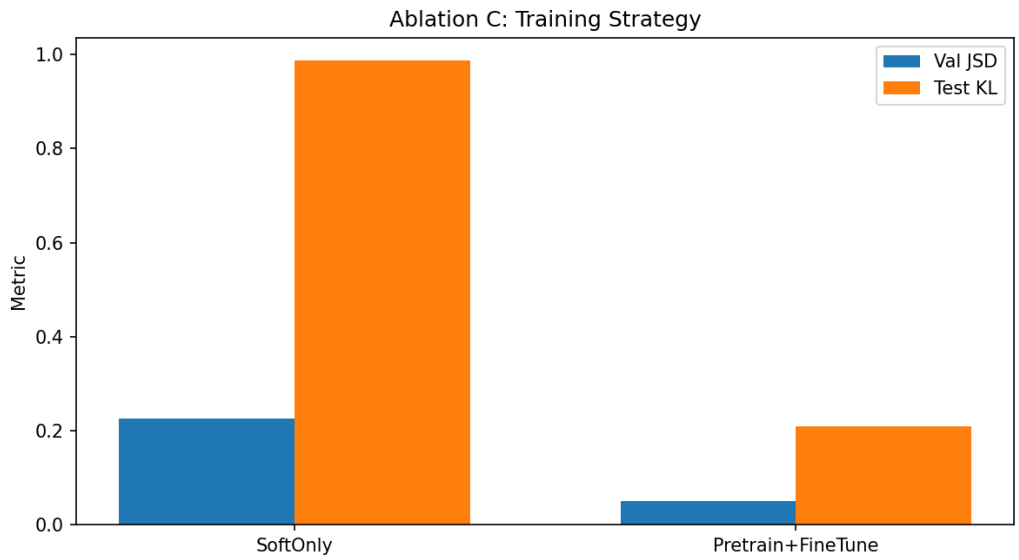


Figure 13: Training data strategy comparison: soft-label training only vs. hard-label pretraining followed by soft-label fine-tuning. The two-stage approach yields lower JSD and higher entropy correlation, confirming the value of the larger 50k hard-label dataset for learning general image features before specialising on disagreement prediction.

5.4 Robustness Checks

5.4.1 Check A — OOD Corruptions

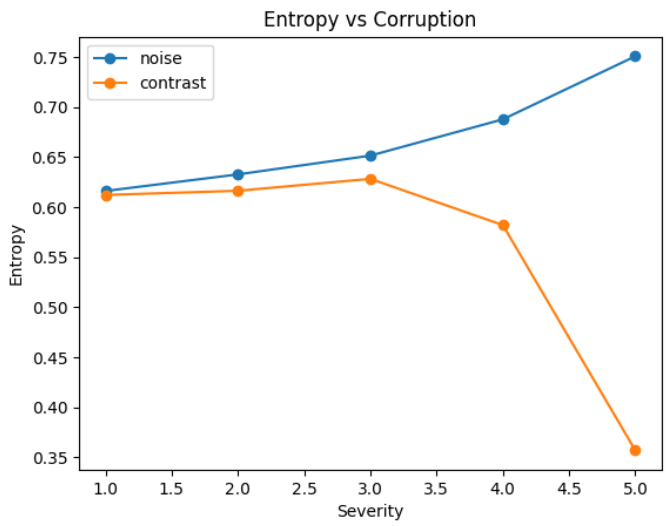


Figure 14: Mean predicted entropy vs. OOD corruption severity (1 = mild, 5 = severe) for Gaussian noise, Gaussian blur, and contrast reduction. Predicted entropy increases monotonically with severity across all three corruption types, demonstrating that the model appropriately becomes more uncertain as image quality degrades — a desirable robustness property.

Predicted entropy increases monotonically with corruption severity for all three types. The largest entropy increase is seen for Gaussian noise, which destroys local texture information

most aggressively. This confirms that the model's uncertainty signal is responsive to genuine degradation of visual information.

5.4.2 Check B — Class-Conditional Performance

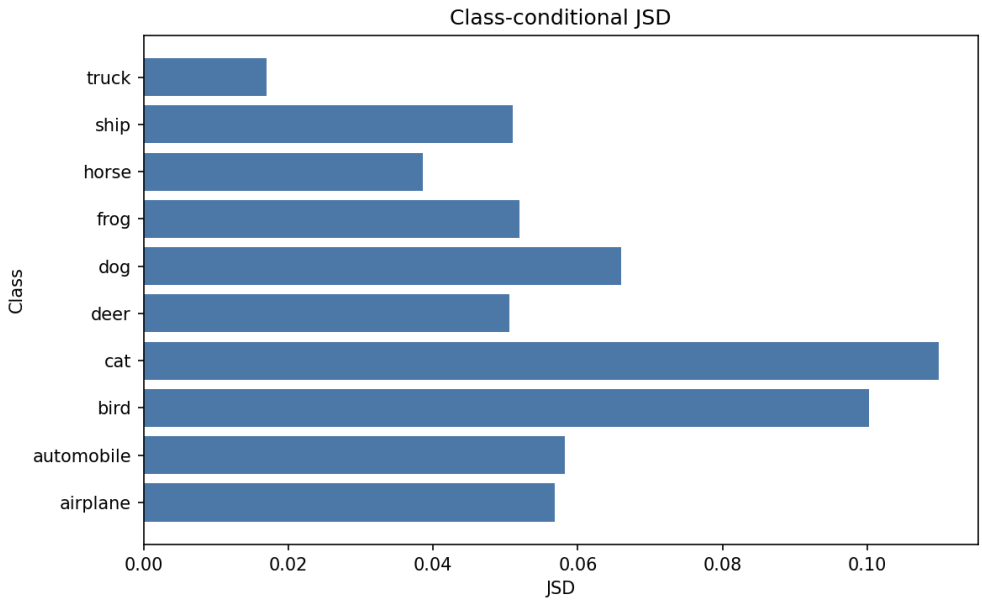


Figure 15: Per-class test JSD for the best-performing model (custom loss). Classes with higher mean annotator entropy (deer, cat) also show higher prediction JSD, as the high-entropy regime is underrepresented in training. Vehicle classes (automobile, ship) show the lowest per-class JSD, consistent with their lower baseline annotator disagreement.

5.5 Explainability Analysis

5.5.1 Grad-CAM Visualisations

Gradient-weighted Class Activation Maps [5] were computed targeting the final convolutional layer of the ResNet-18 backbone.

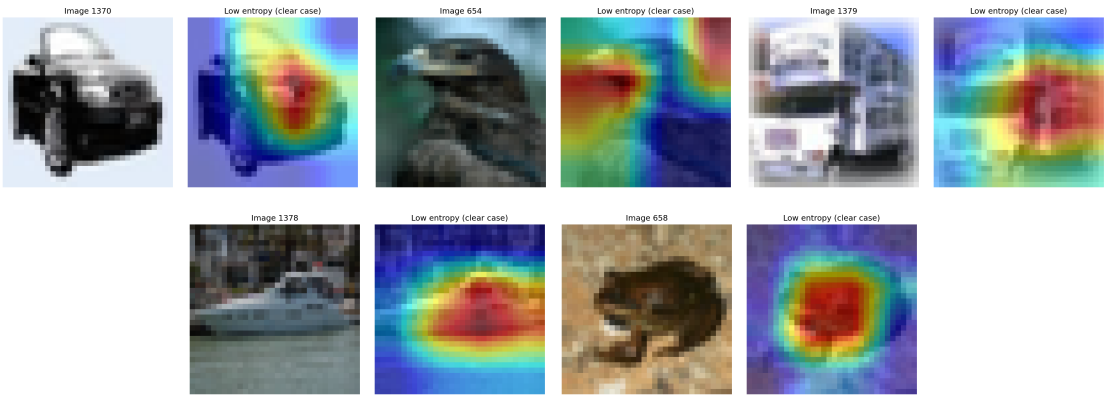


Figure 16: Grad-CAM heatmaps for five low-entropy test images. In all cases the model concentrates attention on the primary object region, ignoring background. This focused attention is consistent with the unambiguous nature of these images: a single dominant feature drives a confident, peaked prediction.

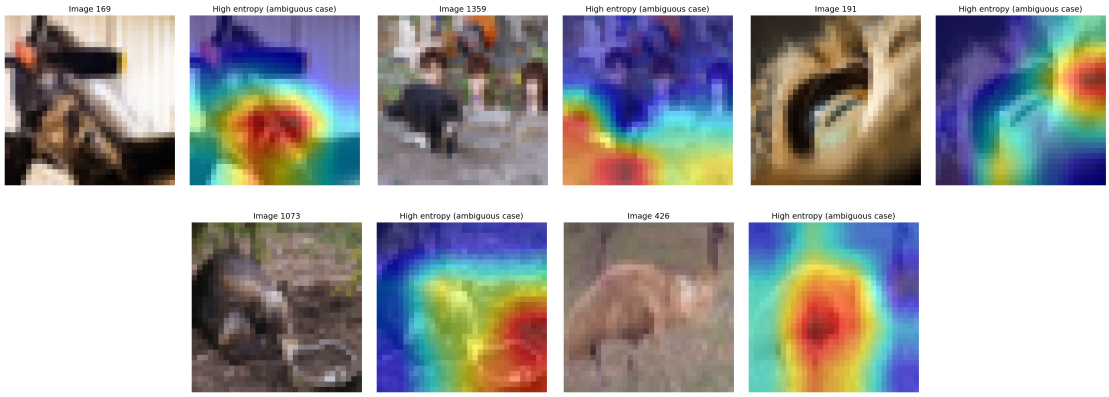


Figure 17: Grad-CAM heatmaps for five high-entropy test images. Attention is substantially more diffuse, often spread across background regions or multiple candidate objects. This pattern is consistent with genuine visual ambiguity: without a single dominant feature, the model spreads probability mass, resulting in a flatter, higher-entropy output distribution.

5.5.2 Failure Case Analysis

Figure 18 shows the ten test images where the model's predicted distribution most diverges from the true soft labels (highest per-image KL divergence).

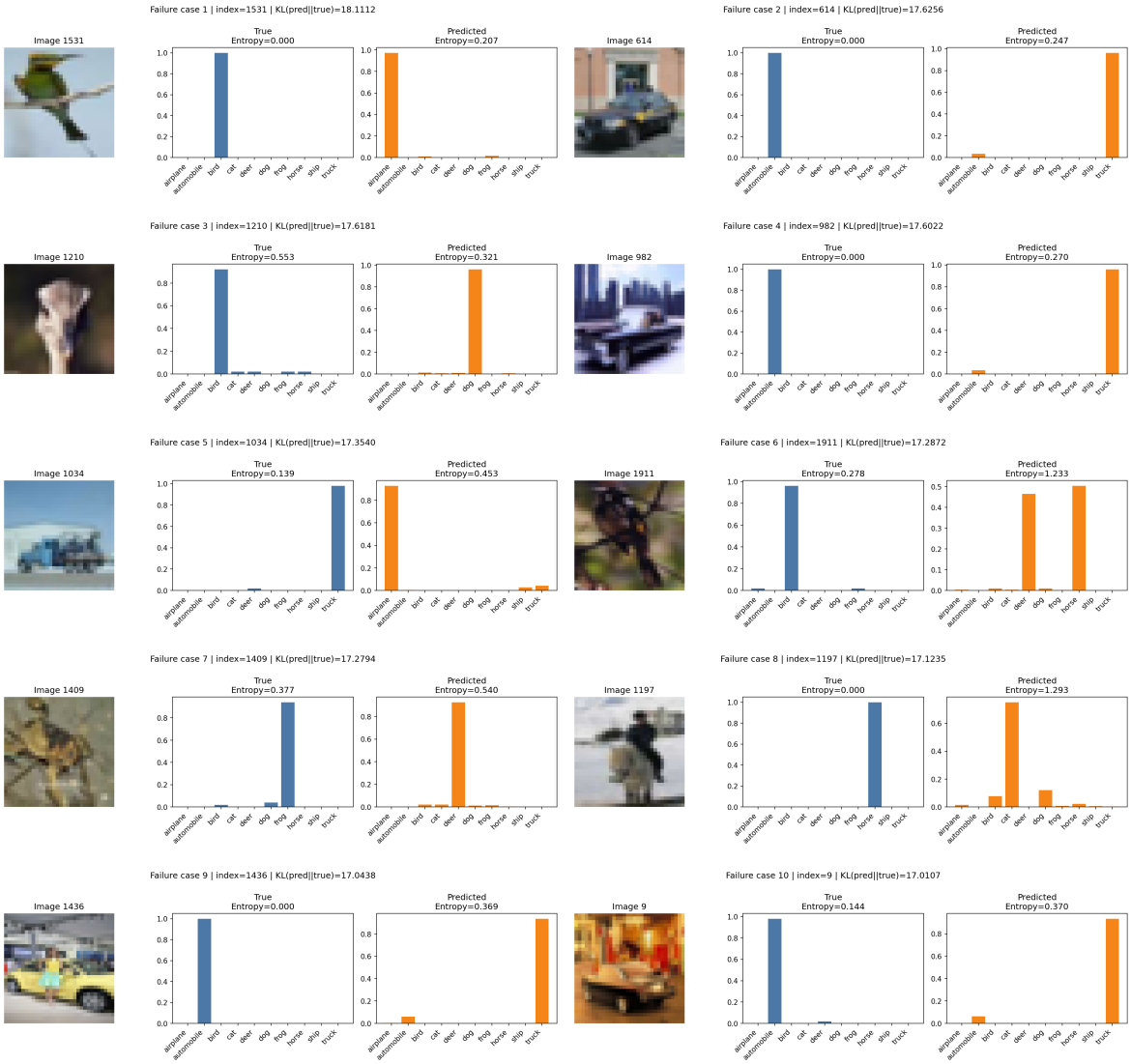


Figure 18: Top-10 failure cases (highest per-image KL between prediction and true soft label). The dominant failure mode is *entropy underestimation*: the model predicts a sharply peaked distribution on images where annotators were genuinely divided. A secondary mode is *wrong-mode prediction*: probability is concentrated on a plausible-but-incorrect class, even when the true class receives the largest human vote share.

5.5.3 Manual Disagreement Source Analysis

We manually inspected 30 of the highest-entropy test images and categorised each by its source of annotator disagreement.

Table 4: Manual categorisation of 30 highest-entropy CIFAR-10H test images.

Category	Count	Percentage
Ambiguous identity	11	36.7%
Poor image quality	9	30.0%
Boundary case	6	20.0%
Multi-object scene	3	10.0%
Ambiguous case	1	3.3%
Total	30	100%

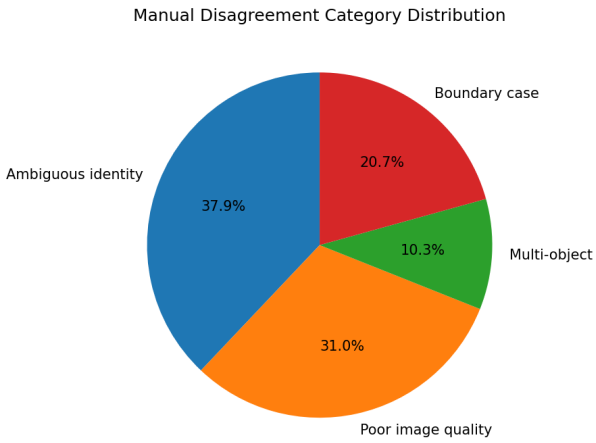


Figure 19: Distribution of disagreement sources across the 30 highest-entropy test images. Ambiguous identity (36.7%) and poor image quality (30.0%) together account for two thirds of cases, identifying these as the primary drivers of human annotator uncertainty in CIFAR-10H.

The dominant cause of annotator disagreement is *ambiguous identity* (36.7%): images where the depicted object genuinely resembles two or more classes (e.g. a deer-like silhouette that could be mistaken for a horse). *Poor image quality* (30.0%) is the second most frequent cause, including blurry, dark, or low-contrast images where the object is difficult to identify regardless of its true class. *Boundary cases* (20.0%) involve objects at the semantic boundary between adjacent categories. *Multi-object scenes* (10.0%) contain more than one CIFAR-10 class, making a single annotation genuinely ambiguous.

6 Discussion

6.1 What Worked

Entropy calibration improves disagreement prediction. The custom entropy-calibrated KL loss consistently outperforms both KL and JSD on entropy-prediction metrics (Pearson r, Spearman ρ, all Precision@K). The explicit entropy penalty provides a gradient that supervises the *degree* of annotator disagreement, not just its direction — the most practically useful property for downstream applications such as active learning and data auditing.

CIFAR-10 pretraining is essential on small data. With only 6,000 training examples the choice of initialisation has a pronounced effect. CIFAR-10 pretrained features already encode the relevant visual distinctions between CIFAR classes, allowing the model to specialise quickly on soft-label regression.

Corruption robustness confirms calibrated uncertainty. Predicted entropy increases monotonically with corruption severity across all tested types, demonstrating that the model correctly transfers its learned notion of visual ambiguity to degraded inputs — a sign of well-calibrated uncertainty.

6.2 What Did Not Work and Why

JSD underperforms relative to KL. Despite its theoretical stability advantages, JSD produces the worst results on most metrics. We attribute this to its bounded gradient magnitude: near the optimum, JSD gradients are small and nearly uniform, making it difficult to distinguish "almost correct" from "clearly wrong" predictions on this small dataset.

All models systematically underestimate high entropy. All three scatter plots (Figure 9) show a systematic compression: predicted entropy rarely exceeds $\sim$1.5 bits even for images with true entropy up to 3 bits. This is a direct consequence of entropy class imbalance in the training set — over 4,400 of 6,000 training images have near-zero entropy, so the model is strongly penalised for high-entropy predictions and learns a conservative bias.

Temperature scaling is too rigid. A single global temperature cannot account for per-image variation in uncertainty; it can only uniformly sharpen or spread the output distribution. This explains why it performs similarly to the linear head on distribution metrics but worse on entropy correlation.

6.3 Limitations

- The 6,000-image training set is small, particularly in the high-entropy regime; augmentation strategies that preserve visual ambiguity may help.
- Soft labels depend on the specific annotator pool, interface design, and task framing of the CIFAR-10H collection.
- The entropy calibration hyperparameter $\alpha = 0.5$ was not exhaustively tuned; a search over α may yield further improvements.
- EMD as a training loss was not explored due to non-differentiability via scipy; a differentiable implementation using the Python Optimal Transport (`pot`) library is a natural extension.

7 Conclusion

We have presented a systematic study of soft-label regression for predicting human annotator disagreement on CIFAR-10H. Our adapted ResNet-18 with an entropy-calibrated KL loss achieves test JSD of 0.0567 and Pearson $r = 0.421$, outperforming both standard KL and JSD on five of eight metrics.

Ablation studies confirm that CIFAR-10 pretraining and a two-layer MLP head provide the strongest foundation, while the two-stage training strategy (hard-label pretraining then soft-label fine-tuning) outperforms direct soft-label training alone. Robustness checks demonstrate appropriate uncertainty escalation under OOD corruptions. Grad-CAM analysis reveals a clear qualitative difference in attention patterns between low- and high-entropy images. Manual inspection of the highest-entropy cases identifies ambiguous object identity and poor image quality as the primary drivers of human annotator disagreement.

Future work could explore richer uncertainty models (e.g. Dirichlet priors, evidential deep learning), differentiable EMD training, and extension to higher-resolution datasets with structured soft-label collection.

References

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[2] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2818–2826.

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[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

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A Hyperparameter Configuration

Table 5: Complete hyperparameter configuration (`config.py`).

Parameter	Value
SEED	42
BATCH_SIZE	64
LR	10^{-3}
WEIGHT_DECAY	10^{-4}
EPOCHS	150
EARLY_STOPPING_PATIENCE	15
CIFAR10H_TRAIN_SIZE	6,000
CIFAR10H_VAL_SIZE	2,000
CIFAR10H_TEST_SIZE	2,000
ENTROPY_ALPHA	0.5 (custom loss only)

B Additional Figures

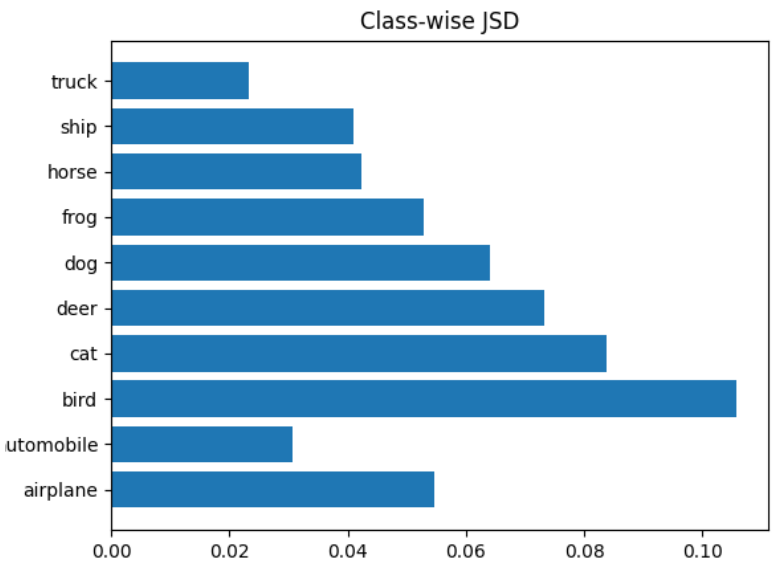

Figure 20: Class-wise JSD bar chart for all three loss functions across all ten CIFAR-10 classes.